STOCK RESEARCHER AGENT

Consolidated Master Guide

Quick Start • Operational Runbook • Infrastructure • Development Framework

Document status	Consolidated working guide
Version / date	v1.0 · 11 October 2026
Repository	appurajacool2015/stock-researcher-agent
Production API	https://stock-researcher-agent.onrender.com
Source documents	Master Manual v2.4; Quick Start; Operational Runbook; Deployment Evidence

Purpose. One place to resume work quickly, verify the known production checkpoint, and find the detailed infrastructure and framework notes without reading four overlapping documents.

Important: This guide consolidates evidence recorded on 11 October 2026. It does not independently verify that the services are live today. Re-run the checks in Part II before relying on the status.

Operating principle

The Render research pipeline is the source of truth for numerical metrics, status tags, and provenance. OpenClaw/Requesty may explain and format the result, but must not invent or independently calculate financial indicators.

How to use this guide

If you are returning after a break, read Parts I and II first. Use Parts III–V only when you need to understand or change the infrastructure, development workflow, or roadmap.

Contents

Part	What it answers
I. Get up to speed	What the project is, what was last observed, and the minimum starting checklist.
II. Verify the system	Commands for Render, Requesty, OpenClaw and the stock-research skill.
III. Infrastructure and architecture	Service responsibilities, request flow, deployment configuration and account references.
IV. Development, deployment and recovery	Local development, Git deployment, tests, troubleshooting and security.
V. Framework and roadmap	ResearchResult, data quality, Supabase persistence plan and open verification items.

Part I — Get up to speed

Project in one sentence

A deterministic stock-research REST API on Render collects and assembles fundamentals, market/technical data and news; OpenClaw on an Oracle Cloud VM is intended to expose that research through chat, using Requesty for language-model inference.

Last documented checkpoint (11 October 2026)

Component	Recorded state	What that does / does not prove
Render REST API	Dashboard showed Live; /health returned healthy.	API health was observed at that checkpoint; recheck now.
Research endpoint	/research/TCS returned ResearchResult v1; overall_status PARTIAL.	A usable response was observed; PARTIAL requires field-level review, not automatic failure.
Requesty inference	OpenClaw model test had previously succeeded.	LLM connectivity was tested separately from the research API.
OpenClaw → Render	Integration must be verified in the skill/tool.	A live Render service does not prove automatic skill invocation.
Supabase persistence	Roadmap / planned.	Do not assume caching or history storage is implemented.

Minimum resume checklist

1. Connect to the existing Oracle Cloud VM using the documented SSH command and confirm the OpenClaw gateway status.
2. Call Render `/health`.
3. Call one research endpoint, for example `/research/TCS`, and inspect the response.
4. Test Requesty independently.
5. Confirm the stock-research skill actually calls Render and handles non-2xx responses.
6. Record the time, deployed commit, HTTP status, overall_status, data-quality summary and any errors.

Part II — Verify the system

A. Render API health

```
curl -i https://stock-researcher-agent.onrender.com/health
```

Expected previously observed body:

```
{"ok":true,"service":"stock-researcher-api","status":"healthy"}
```

B. Research endpoint

```
curl -i https://stock-researcher-agent.onrender.com/research/TCS
```

For a concise INFY check, use this corrected one-line command. It avoids Markdown link syntax and shell line-continuation issues:

```
curl -fsS 'https://stock-researcher-agent.onrender.com/research/INFY' | jq '{symbol: .symbol, status: .overall_status
```

If jq is not installed, retrieve the full JSON instead:

```
curl -fsS 'https://stock-researcher-agent.onrender.com/research/INFY'
```

C. Check Requesty / OpenClaw

```
openclaw --version
systemctl --user status openclaw-gateway --no-pager
openclaw models list
openclaw agent --agent main --message "Reply with exactly: Requesty connection OK." --timeout 60
```

D. Check the skill wiring

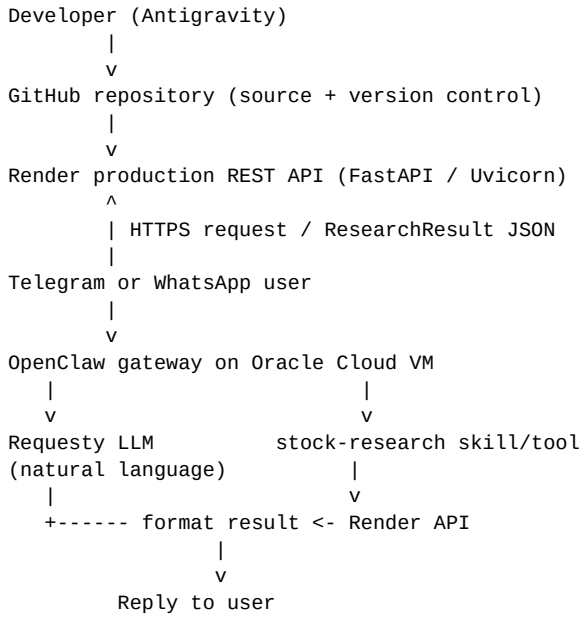
Inspect the stock-research skill/tool implementation and logs. It should call the Render HTTPS endpoint, parse the returned JSON, preserve field-level evidence, handle HTTP errors, and return a clear failure when the API cannot be reached.

Interpretation guardrails

Observation	Interpretation
HTTP 200 + overall_status PARTIAL	The request may have succeeded. Review unavailable/stale/conflicting fields and provenance.
GET / or HEAD / returns 404	Not proof of outage. Use the documented /health endpoint.
Slow first request	The recorded Free instance can spin down; retry and inspect Render logs.
API healthy, chat tool fails	Test Requesty and OpenClaw separately, then verify the skill calls Render.

Part III — Infrastructure and architecture

End-to-end flow



Component responsibilities

Component	Role / boundary
Antigravity	Documented local environment for editing and validating code.
GitHub	Repository and deployment source; not the live market-data API.
Render	Production REST API hosting the FastAPI app and deterministic research pipeline.
Research pipeline / adapters	Collects fundamentals, technical/market data and news; supplies the canonical research result.
Oracle Cloud VM + OpenClaw	Chat gateway and tool orchestration; the documented SSH user is ubuntu.
Requesty	LLM inference route used by OpenClaw; documented model is google/gemma-4-31b-it.
Supabase	Planned PostgreSQL persistence/cache layer; not recorded as live.

Recorded Render configuration

Setting	Recorded value
Service / environment	stock-researcher-agent / Production
Repository / branch	appurajacool2015/stock-researcher-agent / main
Runtime / region	Python 3 / Singapore
Root directory / plan	Repository root / Free instance
Start command	uvicorn interfaces.api:app --host 0.0.0.0 --port $PORT
Health route	/health
Deployment label / commit	added rest api / 7ef7c63 (recorded checkpoint)
Build command	Not visible in the supplied evidence; verify in Render Settings before changing/recreating service.

Service references

Render health: /health
Example research: /research/TCS
FastAPI docs: /docs
Requesty base URL: https://router.requesty.ai/v1

Part IV — Development, deployment and recovery

Normal development path

1. Edit and validate in Antigravity.
2. Run the relevant tests locally.
3. Push approved changes to the GitHub main branch.
4. Confirm the Render deployment and commit.
5. Re-run health and research smoke tests.
6. Test the chat/skill integration end-to-end.

Useful local commands

```
.venv-market/bin/python -m unittest tests/test_api.py
.venv-market/bin/python -m interfaces.api
```

The source manual reported 12 tests passing at the recorded checkpoint. Re-run tests rather than assuming that result still applies.

Deploy/recovery checklist

1. Open the Render service and check the latest deployment status, logs and commit hash.
2. Confirm the Start Command uses $PORT and binds to 0.0.0.0.
3. Confirm the configured Build Command in Render Settings; the supplied screenshot did not show it.
4. Run /health and a representative /research/{symbol} request.
5. Inspect HTTP status, overall_status, data_quality_summary, field statuses, provenance and as_of_date.
6. Verify the OpenClaw gateway and Requesty model separately.
7. Confirm the skill explicitly calls Render and handles timeouts/non-2xx responses.
8. Document the commit and observed results after the test.

Troubleshooting matrix

Symptom	What to check / do
Slow first request or timeout	Retry; inspect Render logs; allow a timeout suitable for cold starts (the skill example uses 75 seconds).
Render build failure	Read full logs and verify the actual Build Command/dependencies; do not guess.
Startup or port failure	Use --host 0.0.0.0 --port $PORT; do not hard-code 10000.
GET / or HEAD / returns 404	Use GET /health for health checks.
Research status PARTIAL	Inspect missing fields, provenance and as-of dates; PARTIAL is not automatically a pipeline failure.
Requesty/model failure	Check gateway, model list and REQUESTY_API_KEY presence without printing the value.
OpenClaw does not invoke research	Inspect the skill/tool wiring and execution logs.

Security rules

Never print, commit, or paste the contents of `~/.openclaw/.env` or any API key. Keep secrets out of Git, logs, screenshots, documentation, and chat. Account emails/labels identify logins; they are not credentials. Keep service-role database credentials on private backend workers only.

Part V — Framework details and roadmap

Canonical research payload

The API returns a ResearchResult payload (the supplied evidence includes schema_version 1.0). The pipeline owns financial values, status tags, data-quality counts and provenance. The LLM layer should summarize and format the payload, not manufacture metrics.

The recorded TCS sample

Field	Recorded sample value
Symbol / exchange	TCS / NSE
Company	Tata Consultancy Services Ltd
Symbol match confidence	EXACT
Overall status	PARTIAL
Data quality summary	25 verified · 4 derived · 2 unavailable · 0 stale · 0 conflicting · 0 failed

These figures are historical evidence from the supplied documents, not current market data or a guarantee of future responses. Always inspect field-level provenance and as_of_date.

Planned Supabase persistence — not yet verified as implemented

The Master Manual describes a proposed PostgreSQL table named stock_research_runs to store research snapshots and data-quality summaries. The proposed read-cache flow checks for a result for the same symbol from the last six hours; on a cache miss, it calls Render, persists the payload and returns the result.

```
CREATE TABLE IF NOT EXISTS stock_research_runs (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  symbol VARCHAR(20) NOT NULL,
  exchange VARCHAR(10) DEFAULT 'NSE',
  company_name TEXT,
  overall_status VARCHAR(20) NOT NULL,
  data_quality_summary JSONB NOT NULL,
  result_payload JSONB NOT NULL,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

The original manual also proposes indexes on symbol/created_at and overall_status, Row Level Security, and protected SUPABASE_URL / SUPABASE_KEY configuration. Treat this as a roadmap until implementation and connectivity are verified.

Open items to verify on the next session

Priority	Verification item
High	Confirm OpenClaw stock-research skill invokes the Render API end-to-end.
High	Re-run Render health and research smoke tests; record current status and commit.
High	Confirm actual Render Build Command and current environment settings.
Medium	Re-run local unit tests and record current result.
Medium	Confirm whether Supabase schema/cache is still only planned or has since been implemented.
Medium	Review account references and remove any unnecessary personal identifiers from shared copies.

Document consolidation notes

This guide intentionally combines overlapping content rather than appending all source documents page-for-page. The source files had repeated account directories and architecture addenda, differing version labels, and a deployment-evidence section that repeated the runbook. This v1.0 guide uses the 11 October 2026 checkpoint and preserves uncertainty where the supplied evidence was incomplete.

Version note: the source Master Manual filename says v2.4, while its cover/body still labels the manual v2.2 and the appended correction says v2.4. The Operational Runbook filename says v2.2 but its internal title says v2.1. The consolidated guide avoids implying those internal version labels are fully reconciled.